Experiment No. 9: Image Scaling Techniques

Aim: To implement image scaling techniques in Python using OpenCV to resize an image to both larger and smaller sizes.

Program (Google Colab):

```
import cv2

import matplotlib.pyplot as plt

from google.colab import files


# Upload an image

uploaded = files.upload()

image_path = next(iter(uploaded))


# Read the image

img = cv2.imread(image_path)

img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)


# Get original dimensions

h, w = img.shape[:2]


# Resize to a bigger size

big = cv2.resize(img, (w * 2, h * 2),

                 interpolation=cv2.INTER_CUBIC)


# Resize to a smaller size

small = cv2.resize(img, (w // 2, h // 2),

                   interpolation=cv2.INTER_AREA)
```

```
# Display images
plt.figure(figsize=(12, 5))

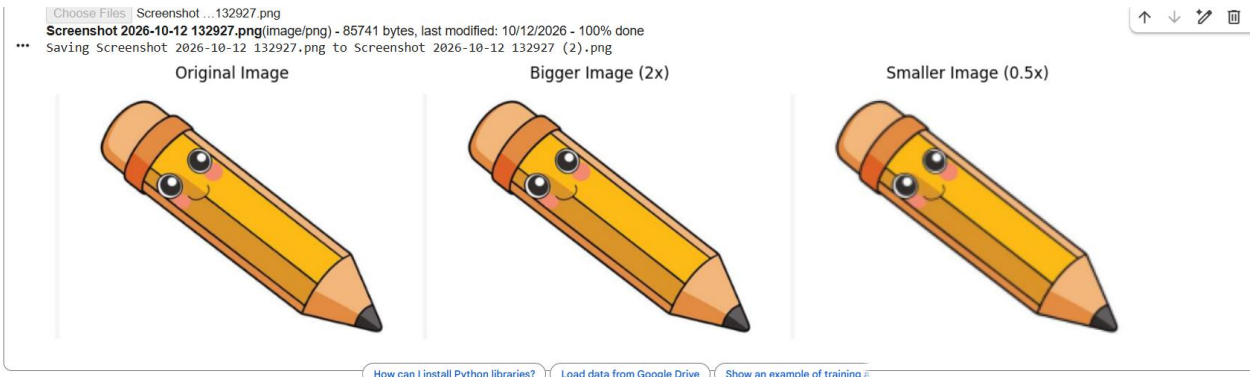
plt.subplot(1, 3, 1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(big)
plt.title("Bigger Image (2x)")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(small)
plt.title("Smaller Image (0.5x)")
plt.axis("off")

plt.tight_layout()
plt.show()
```

OUTPUT:



Expected Output:

1. Original image.
2. Enlarged image with twice the original width and height.
3. Reduced image with half the original width and height.

Result: Image scaling was successfully implemented using Python and OpenCV. The image was resized to both larger and smaller dimensions using cv2.resize().